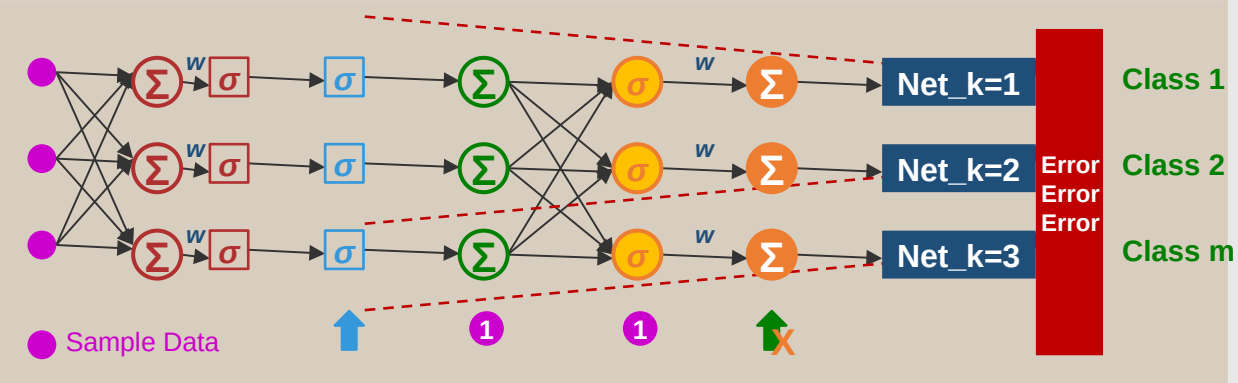
W_{ji} *Affects* Net_j $\Rightarrow$

Net_j *Affects* O_j $\Rightarrow$

O_j *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ $\Rightarrow$

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*



$$\frac{\partial E}{\partial w_{ji}} = \left[\frac{\partial E}{\partial Net_{k_1}} \cdot \frac{\partial Net_{k_1}}{\partial O_j} + \frac{\partial E}{\partial Net_{k_2}} \cdot \frac{\partial Net_{k_2}}{\partial O_j} + \cdots + \frac{\partial E}{\partial Net_{k_m}} \cdot \frac{\partial Net_{k_m}}{\partial O_j}\right] \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial Net_k} \Rightarrow -\delta_k$$

$$\text{Net}_k = \sum_k w_{kj} O_j \Rightarrow \frac{\partial \text{Net}_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1+e^{-Net_j}} \Rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1-O_j)$$

$$\text{Net}_j = \sum_j w_{ji} O_i \Rightarrow \frac{\partial \text{Net}_j}{\partial w_{ji}} = O_i$$